

Reproductive System Disorders

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I introduced some notable aspects of the anatomy of the reproductive tracts in small mammals in the [anatomy lecture](#). Please review the relevant information prior to reading this. In particular, recall the information on the anatomy of the uterus in rabbits, the location of the urethral opening in female rodents, the anatomy of the penis in male guinea pigs, or the extensiveness of the mammary tissue in female rats. I did not discuss much the reproductive biology of the small companion mammals. Below are some useful information on reproductive biology normals in the most common species seen as pets.

	Ferret	Rabbit	Guinea pig	Chinchilla	Rat
Puberty	4-8 months	4-9 months	2-4 months	8 months	10-12 weeks
Ovulation	Induced	Induced	Spontaneous	Spontaneous	Spontaneous
Estrous cycle	Estrous until bred	No regular estrous cycle	15-17 days	25-50 days	4-5 days
Gestation time	39-42 days	31-32 days	69 days	105-118 days	21-22 days
Average litter size	8	7-9	2-5	1-6, typically 2	8-16

Before we go further, I would like to give you some important fun facts of high clinical relevance, which highlight a peculiar feature of each species/group of species that we see (and 1 in naked mole rat, just because we don't talk enough about them).

-**Ferret:** unspayed females will go into **continuous estrous, which will ultimately lead to hyperestrogenism**, in turn leading to aplastic anemia.

-**Rabbits:** females pull hair from their gigantic dewlap and other parts of their body to line their nest prior to kindling.

-**Guinea pigs:** if sows are not bred before 7-8 months old, they will have an **incomplete separation of pubic bones that will lead to dystocia** at subsequent partu-

rition.

-**Rodents:** after copulation, there is a **vaginal (copulatory) plug**, which will fall off after a few hours. It is hard and rubbery.

- **Naked mole rat:** this ugly (but weirdly cute) species has a matriarchal eusocial behavioral structure whereby there is a single breeding female (the Queen naked mole rat) even though colonies can have hundreds of individuals and there are a few breeding males. The Queen may produce over 1,000 offsprings.

Also naked mole rats live super long for a rodent and don't really get [cancer](#) (but looking at them, there was clearly a price to pay for that superpower).



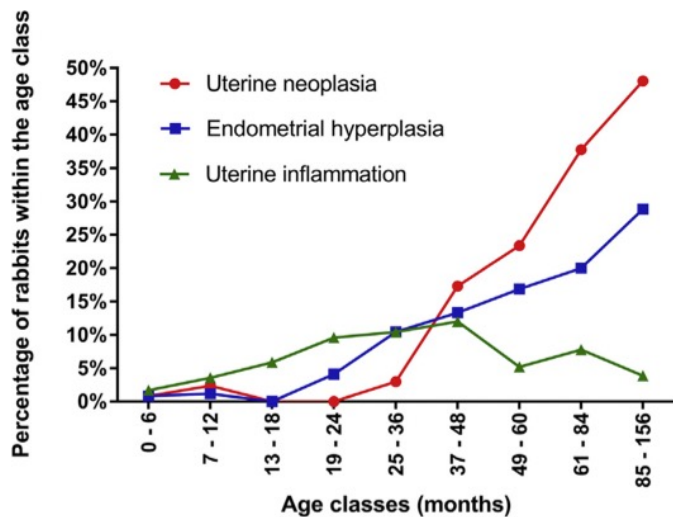
We will review only common reproductive diseases or what is of particular significance in a given small mammal species. We will spend more time on rabbits and guinea pigs as they have a high prevalence of reproductive disorders. For instance, it has been estimated at 25-30% in rabbits on post-mortem studies. Also, most ferrets are already spayed prior to acquisition, so you will rarely deal with uncastrated/unspayed ferrets. We will therefore mainly focus on reproductive diseases of the sterilized ferret. Lastly, we won't cover dystocia as you won't deal with them very often unless you deal with breeders. The management of dystocia is very similar to other mammals anyway, and it often ends up in C-section. We won't cover perinatology as well in this lecture.

UTERINE DISEASES

Uterine Neoplasia in Rabbits

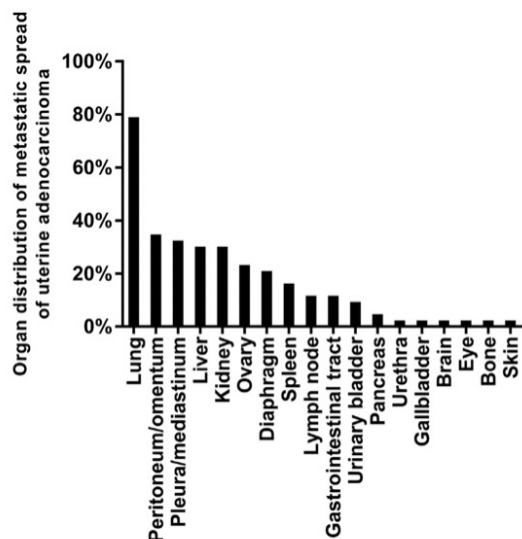
Uterine adenocarcinoma is, by far, the most common cancer of rabbits and the most common repro-

ductive disease. The main risk factor is age in unspayed does. With age, the endometrium undergoes progressive changes with decrease in cellularity and increase in collagen content. It is postulated that tumors are associated with senile atrophy of the endometrium. **The prevalence can be at a staggering 50-70% in rabbits over 5-7 years old.** It is so common, that you should always rule it out in older unspayed does with any presentation. Look at the graph below taken from a retrospective post-mortem study on rabbits.



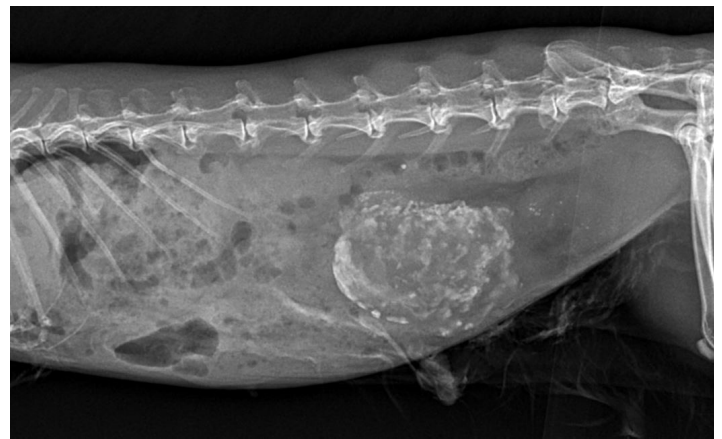
Other less common uterine neoplasia to keep on the differential for uterine masses include adenoma, leiomyoma, and leiomyosarcoma.

Uterine adenocarcinoma are **slowly developing tumors with an early local invasion** that can be quite extensive. **After 1-2 years of growth, it eventually metastasizes, primarily to the lungs and peritoneum/mediastinum.** On post-mortem studies, about half of the rabbits were found to have metastasis, so there is a **high rate of metastasis** with this cancer.

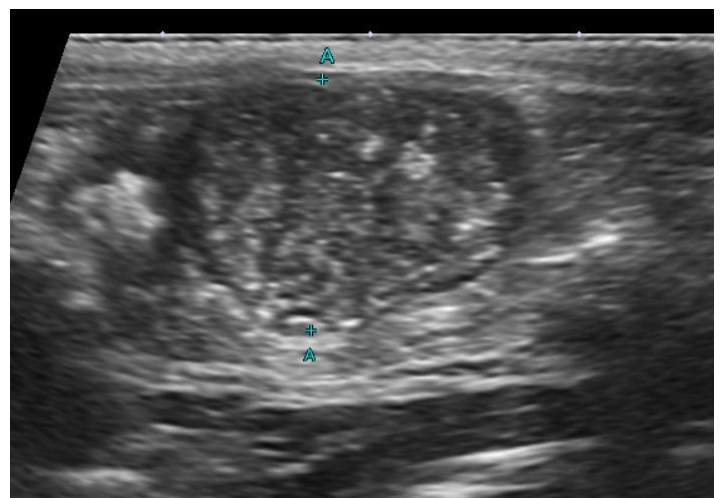


Clinical signs are often non-specific and include pain, gastrointestinal stasis, anorexia, and lethargy. In some cases, a **serosanguinous vulvar discharge is seen, but this is inconsistent.** These tumors are also associated with **cystic mammary swelling and cystic ovaries.** On abdominal palpation, **a caudal abdominal mass can typically be palpated.** Rabbits may also present dyspneic if significant pulmonary metastasis are present.

The diagnosis mainly relies on diagnostic imaging. **The tumors and metastasis are frequently mineralized** so if you see large mineralized masses in the caudal abdomen in an unspayed doe on radiographs, there is a good chance that it is uterine adenocarcinoma (abscess is a differential for a calcified mass in a rabbit).



Ultrasound is generally more sensitive than radiographs and more useful to differentiate from other causes of uterine enlargement. Ultrasound will also confirm the location of the mass in the uterus and may allow the collection of fine-needle aspirates of that mass for cytologic confirmation.



The tumors are frequently located in both horns and are multicentric. In any case, **staging is recommended** as the rate of metastasis is high so a combination of

whole-body radiographs (or chest radiographs) and abdominal ultrasound is recommended. CT is more sensitive than radiographs for the detection of pulmonary metastasis.

The treatment depends on the staging and the presence of metastasis. **If the diagnosis is early and no metastasis could be detected, then an ovariohysterectomy may be curative** and a median survival time of 2 years has been reported. Rabbits should still be monitored for undetected metastasis at time of surgery. **When metastasis are already present, the prognosis is very poor** and chemotherapy could be considered (not a lot of data on that).

Because the prevalence is so high and the cancer so aggressive, it is **best to spay rabbits early**. If rabbits are young (<6 months), you could get around by only doing an ovariectomy, but most people still do an ovariohysterectomy (the main issue is with spay laparoscopic spays as you can only remove the ovaries with this technique). For older animals, a full ovariohysterectomy needs to be done making sure you remove all the uterus and the most rostral part of the vagina (after the cervixes, see [surgery lecture](#)). **Stump adenocarcinoma** has been reported when the uterus was not fully removed.

Endometrial Hyperplasia in Rabbits

If you look at the prevalence graph on the previous page, you'll see that endometrial hyperplasia is also a very common uterine disorder in rabbits. It seems to be concurrent to uterine adenocarcinoma in some cases and maybe predispose to it (the relationship between the 2 is still unclear). In rabbits, the main form that is seen is **cystic glandular endometrial hyperplasia**, but papillary or mixed forms may also be diagnosed.

The disease is **most commonly subclinical** and diagnosed incidentally on ovariohysterectomy or abdominal ultrasound examination. When clinical signs are present, a serosanguinous vulvar discharge may be seen. **On abdominal palpation, the uterus may feel large and firm. Mammary glands may also appear cystic.**

The diagnostic technique of choice is really an abdominal ultrasound examination where the uterus walls look thickened with multifocal hypoechoic cystic structures throughout the endometrium and no discernible lumen. Some small mineralization throughout may also be seen.



Other Uterine Disorders of Rabbits

Other diseases of the uterus that are not too uncommon, some of which may cause a caudal abdominal mass on palpation, include the following:

- **Uterine polyps**
- **Pyometra, endometritis, and uterine abscess:** these cases frequently present with vaginal discharge. Bacteria commonly implicated include *Pasteurella multocida*, *Staphylococcus aureus* and others.
- **Hydrometra and mucometra:** accumulation of watery fluid in the uterus. These cases will present with abdominal distention. The etiology is unclear.

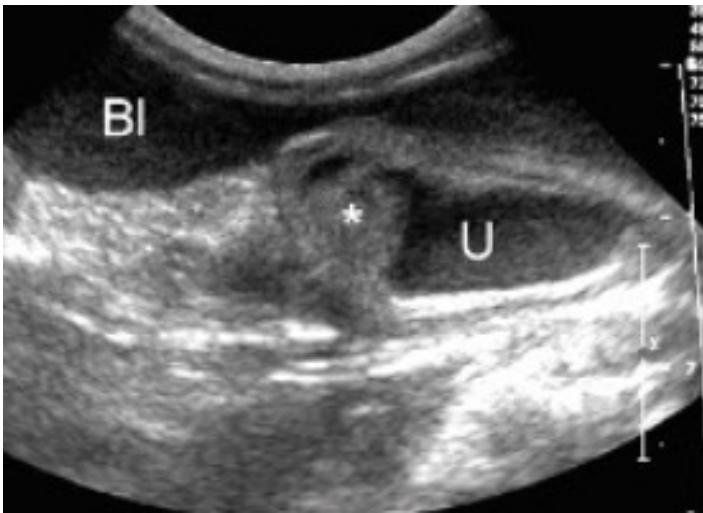


- **Uterine torsion:** more often seen in multiparous breeding females. Does may present shocky or die acutely.

- **Vaginal prolapse:** remember the length of the vagina in rabbits, so the prolapse can be pretty long.

A disease that warrants a special mention is **endometrial venous aneurysm**. This curious disease leads to blood-filled endometrial varices that periodically rup-

ture. **Rabbits present with significant, often very large episodic bleeding including blood clots.** Fatal exsanguination is possible and a low PCV is also frequently seen on hematology. It seems to preferentially affect **young does of large breeds.** It can be found incidentally following a spay. This disease is easily diagnosed by ultrasound, which will show blood filled spaces and blood clots.



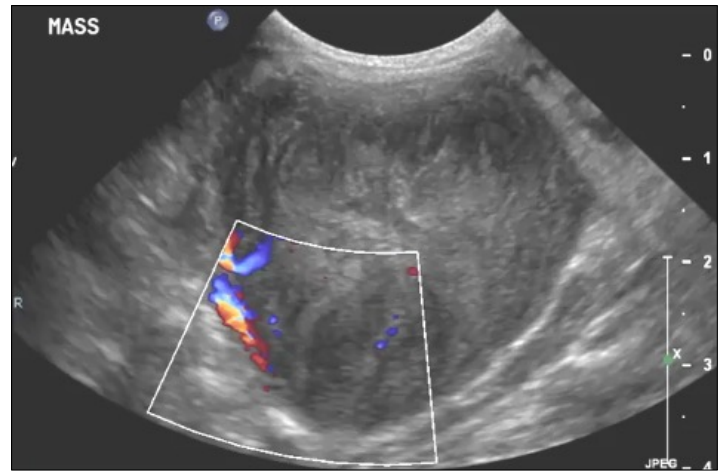
The cause of endometrial venous aneurysm is unknown, but congenital defects in the adventitia of uterine veins are suspected.

The treatment to all these diseases is ovariectomy. Don't forget to submit the tissues for histopathology.

Uterine Diseases of Guinea Pigs

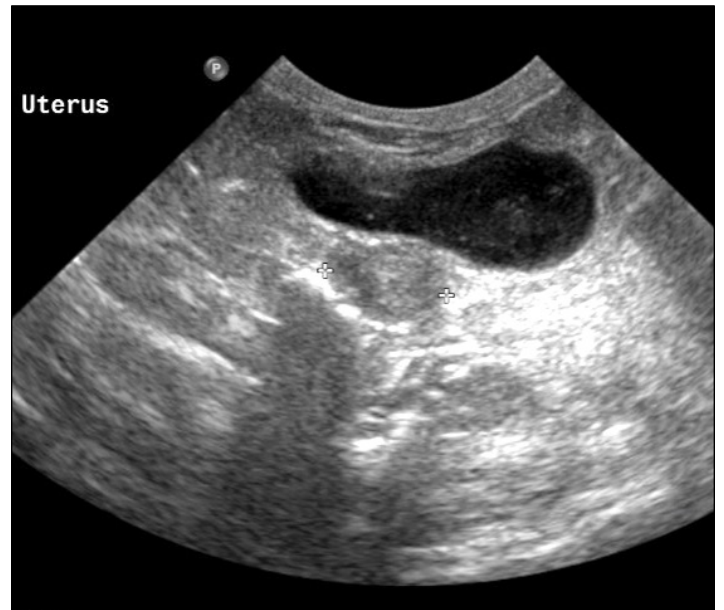
Guinea pigs get uterine tumors too, but it tends to be more benign in them. The most common is **leiomyoma**. They also get adenoma and less commonly malignant tumors such as leiomyosarcoma and adenocarcinoma. The clinical signs include **hemorrhagic vulvar discharge**. Interestingly, some of those have been **associated with the presence of ovarian cysts** (mainly the follicular type, see later). As for most uterine disorders, abdominal ultrasound is the imaging of choice. Below is an ultrasound with color doppler to demonstrate vas-

cularization of a leiomyoma in a guinea pig.



Cytologic samples can be obtained following ultrasound. Ovariectomy is the treatment of choice.

Cystic endometrial hyperplasia is also pretty common in guinea pigs with an approximate prevalence of 15% in older sows. The management is similar as in rabbits including ultrasound for diagnosis and ovariectomy for treatment. Below is an ultrasound of the uterus of a guinea pig with cystic endometrial hyperplasia (structure below the bladder).



Inflammatory conditions of the uterus in guinea pigs include endometritis and pyometra. It tends to be less common than other uterine diseases in this species. The bacteria commonly cultured from these lesions are *Bordetella bronchiseptica* and *Streptococcus* spp.. If surgery is not possible, a possible palliative treatment option is aglepristone.

Ovarian hormone-producing cysts may predispose to these uterine disorders.

OVARIAN DISEASES

Ovarian Cysts in Guinea Pigs

Ovarian cysts are the main reproductive disease of guinea pigs, so you should pay special attention to this section. Over 90% of females will have cysts after 2 years old.



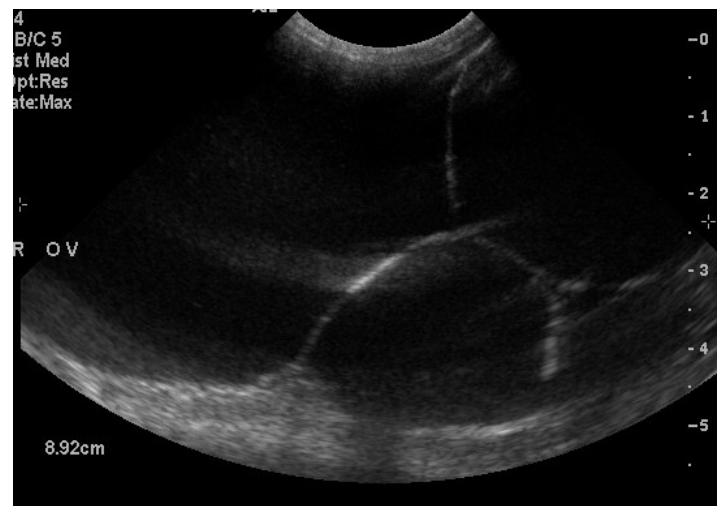
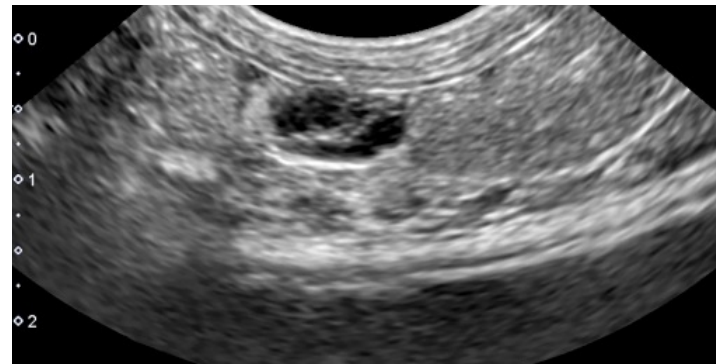
There are essentially two types of cysts.

-Rete ovarii cysts (serous cysts): those are the most common ones with 60-90% of all cysts being serous cysts. The rete ovarii is a group of anastomosing tubules lined by epithelium located at the hilus of the ovary, where vessels and nerves enter the ovary. It is homologous to the rete testis in males. These cysts are often bilateral and, curiously, when unilateral tend to affect the right more than the left (supposedly because of the size of the cecum). **The number of cysts and their size increase with age.** There seems to be an association between these cysts and the presence of uterine diseases (see before). These cysts do not produce hormones, so they are mainly subclinical. When they get big enough, they will lead to pathologic effects such as abdominal distension, weight loss, hunched posture, abdominal pain, anorexia, and decreased fertility. On physical examination, an abdominal mass may be palpated in the area of the ovaries.

-Follicular (functional) cysts: they only constitutes

about 10-20% of all cysts in guinea pigs. Contrary to serous cysts, these cysts **produce hormones and may lead to hyperestrogenism.** Signs that are seen in guinea pigs with follicular cysts include **bilateral flank alopecia, increased clitoris size, hyperkeratosis/hyperpigmentation of nipples, vulvar discharge, and aggressive behavior.**

The diagnosis is pretty straightforward with abdominal ultrasound. See below ultrasound images of a small and a ginormous cyst in two guinea pigs.



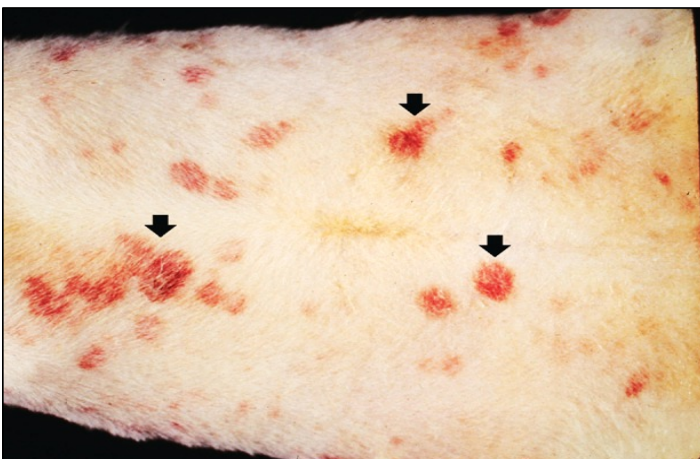
Treatment includes surgical and medical options. **Ovariectomy is the treatment of choice.** With that said, ovariectomy is pretty challenging in guinea pigs when going midline because of the size of the GI and the difficulties to exteriorize ovaries. **Cyst aspiration does not really work in the long term.** It may give some temporary relief, especially for big cysts associated with compression of adjacent structures. However, fluid will reaccumulate with days to weeks and there is always a risk of cyst rupture. **For follicular cysts, medical treatments may be tried to induce luteinization of the cysts with drugs such as injectable GnRH or hCG.** Deslorelin implants have not been shown to work well and may be associated with an alteration of normal estrous and reproductive disorders (other ovari-

an cysts, endometrial cystic hyperplasia, adenomyosis, uterine cervix hyperplasia). Medical treatment does not work on serous cysts (most of the cases unfortunately, but they are also mostly subclinical).

Ovarian Diseases in Ferrets

Since ferrets are induced ovaluators, **jills can stay in persistent estrous if no mating occurs and lead to hyperestrogenism**. Pretty much all ferrets are spayed in North America (and ferrets are banned in California) so you won't see this condition very often, but it is an important disease of ferrets that you still need to know about.

The **prevalence is higher during the reproductive season**. Hyperestrogenism that occurs for a long period of time (such as 1 month) will lead to secondary bone marrow toxicity in ferrets and cause **aplastic anemia and thrombocytopenia** (and pretty much pancytopenia). Clinically, this could be detected by the presence of pale mucus membranes and petechiae and pathologic hemorrhages due to coagulopathies. Significant vulvar enlargement is also seen.



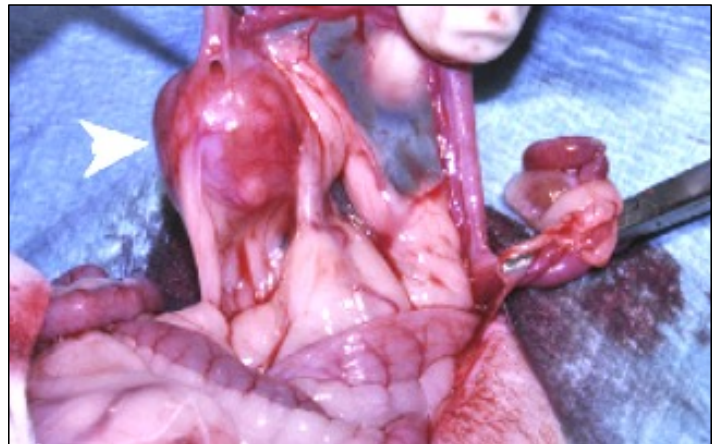
Secondary reproductive diseases such as pyometra or secondary infection (due to decreased WBC) such

as pneumonia can be seen.

The obvious treatment is to spay the ferrets. However, they may be significantly anemic to the point of requiring a **transfusion** (luckily, ferrets do not have blood groups, so any donor will do). If the spay cannot be done on an unstable patient, then medical treatment options include, like for ovarian cysts, **injections of GnRH and hCG to induce ovulation**. Another option is to use vasectomized males for mating to induce ovulation. Jills should not be allowed to stay in estrous for more than a month.

A disease that you may actually see more in spayed jills is **ovarian remnant syndrome**. This happens when there is incomplete removal of the ovary during early spay. The risks are higher when spaying very very young ferrets (<8 weeks) apparently due to inadvertent loss of the ovary in the abdominal cavity. In spayed female ferrets, this is the **main differential diagnosis for adrenal gland disease**. The big difference is when the disease is seen. When signs occur in a ferret less than 3 years old, it is most likely going to be caused by an ovarian remnant; if the ferret is older then adrenal gland disease is more probable. Clinical signs will be those associated with estrogen production and may lead to **hyperestrogenism, so things like anemia, alopecia, vulvar enlargement can be expected**. The diagnosis is based on abdominal ultrasound examination, **adrenal hormone panel (especially if only estradiol is elevated), and exploratory laparotomy**.

Ovarian neoplasias are uncommon overall, but are still the most common reproductive tumors in ferrets. Often, they result from the tumorization of an ovarian remnant. Ovarian tumors seen in ferrets are mainly leiomyoma and granulosa cell tumor, but many other types have been reported.



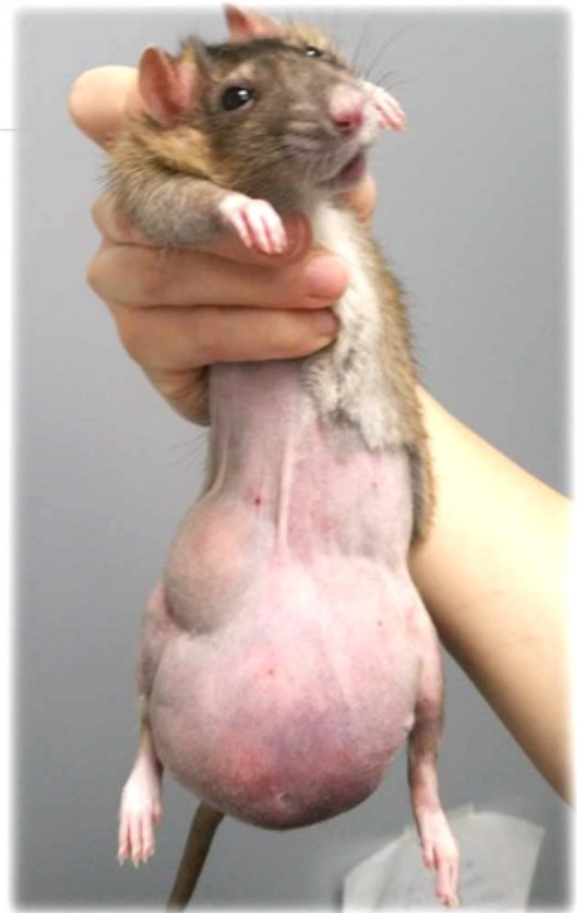
What About Gerbils?

Even if gerbils are illegal in California, it is worth

mentioning that gerbils have a **very high prevalence of ovarian cysts**. As a matter of fact, they occur in 50-80% of older gerbils (and also in hamsters). These are **frequently associated with ovarian neoplasia, in particular granulosa cell tumors**. These cysts are **typically follicular cysts, unlike in guinea pigs, meaning that they produce hormones**. Therefore, clinical signs will commonly include bilateral flank alopecia. Other signs are more associated with the size of cysts in the abdomen. Ovarian cysts are the most common cause of decreased fertility in breeding aged gerbils.



Good news! **Most of these tumors are benign, about 75% of them are fibroadenoma**. The other 25% of tumors are malignant such as carcinomas and fibrosarcomas among other types. They present as typically a (large) masses almost anywhere on the body. Remember that these pesky rats have an **extensive mammary tissue composed of 6 pairs of mammary glands extending from the neck to the tail base**. These masses are slow growing, but most owners will wait until they are fairly big. Their size can be quite amazing and may lead to mobility issues. In addition, the mass may ulcerate and become secondarily infected.



MAMMARY GLAND TUMORS

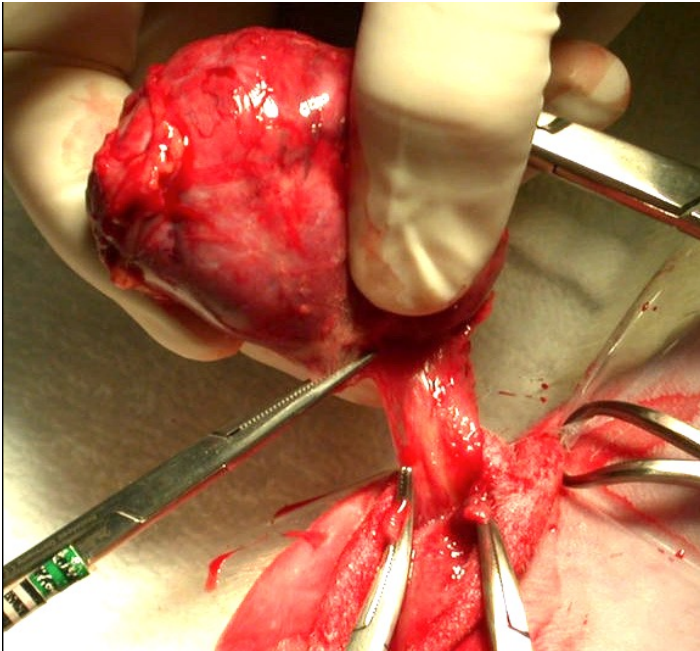
Rats

The main non-infectious disease of rats is mammary gland tumors in females. **The prevalence is high at around 30-70%. There is a possible association with pituitary tumors** (producing prolactin, which promotes mammary gland hyperplasia) as these are frequently found along with mammary gland tumors, but I think whether this association is really an issue or not is controversial. These masses are so common that you should consider **any mass on a rat as mammary gland tumors until proven otherwise**.

The diagnosis is mainly based on the physical examination, but the only way to know whether the mass is benign or malignant is to do histopathology. **Cytology is not super useful for those masses**. The reason is that cytology cannot tell well whether the tumor is benign or malignant because criteria of malignancy for this tumor type is better assessed based on histopathological criteria (invasion into surrounding tissues and vessels, architecture...). So no need to do an FNA.

The treatment mainly consists of **surgical removal** (see [surgical lecture](#)). The surgical excision is curative if the mass was benign. However, **recurrence on the**

same site or another site is common, typically 4-5 months later.



It is best to **spay rats early** to prevent the occurrence of these tumors, preferentially at 3 months of age (see [surgical lecture](#)). If ovariectomy is performed later, the reduction of prevalence will not be as dramatic. A deslorelin implant in place of a spay once a year may be a reasonable alternative. **Secondary prevention after removal of a mass is not effective** as far as we know. So ovariectomy or the placement of a deslorelin implant will not work. Some people also use cabergolin to prevent recurrences.

Non-Rats

In other species of small mammals, mammary gland tumors tend to be more aggressive. **In rabbits, 90% of tumors are carcinomas** (malignant). **In guinea pigs, more than half of the tumors are malignant and also occur in males.** Likewise, **in hamsters, about half are malignant. In mice, most mammary gland tumors are malignant, at about 90% of all tumors.** Not only are they common, they are also frequently multicentric. Adenocarcinoma and fibrosarcoma are the most common and metastasis to the lungs occur. **The mouse mammary tumor virus is a retrovirus** (both exogenous and endogenous) that has a high prevalence. The endogenous virus presents as a provirus in the genome of many strains of laboratory mice.

TESTICULAR DISEASES

Testicular tumors occur from time to time in rab-

bits and are mostly benign. **Granular cell tumors are the most common (and were previously misdiagnosed as interstitial cell tumors).** Cellular origin is undetermined, but most likely Schwann cells. Other tumors that are seen include Sertoli cell tumors and seminomas. The number one clinical sign is a unilateral increased size testis. The treatment of choice is to neuter them. The prognosis is generally good, but is guarded when metastasis are detected (rare).

Other testicular diseases of rabbits include inguinal hernia, cryptorchidism, and orchitis. **Because rabbits have an open inguinal canal, they are prone to inguinal hernia, with the bladder being commonly involved.** Since rabbits can retract their testicles intra-abdominally, you need to make sure the animal is in fact cryptorchid. You may place digital pressure over the caudal abdomen to attempt to push the testicle back out through the inguina canal. **Orchitis presents as a swollen and red testis.** Bacteria that should be kept in mind with this are *Pasteurella multocida* and *Treponema paraluis-cuniculi* (see later) among others. Male rabbits also have the tendency to bite other male entire rabbits on the testicles, which may result in nasty wounds. In those cases, castration may need to be associated with scrotal ablation.

PERINEAL / PENILE DISEASES

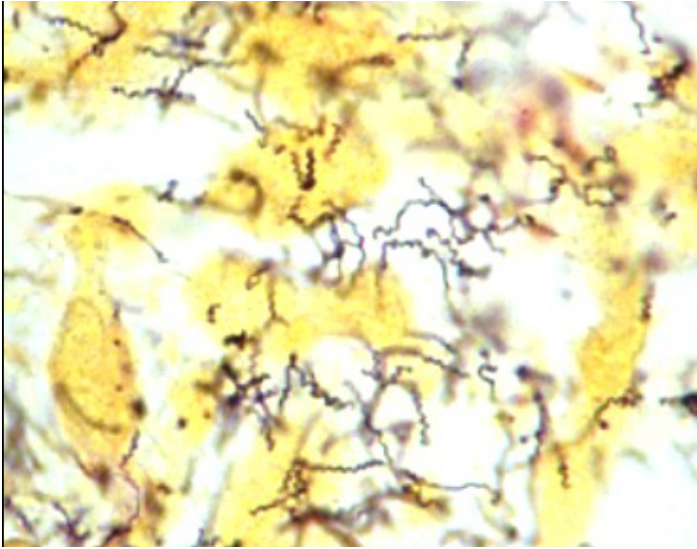
Rabbit Syphilis

Veneral spirochetosis, or rabbit syphilis, is caused by *Treponema paraluis-cuniculi*. It is **non-zoonotic**. It is directly transmitted from individuals to individuals during mating. **Signs are non-specific and consist of redness, edema, vesicles, ulcers, scabs, and local pain over the skin of the perineum and the genitalia.** Rabbits also commonly present (most common reasons of presentation) for lesions on their nose or lips due to auto-infection.



This combination of lesions is almost pathognomonic for rabbit syphilis. The differential diagnosis should still include bacterial dermatitis, dermatophytosis, and ectoparasites.

To confirm the diagnosis, you can perform a biopsy or cytology. **Silver is the stain of choice for spirochetes.**



Wet mount of scabs or skin scrapings can be viewed on dark field microscopy as well. Serology and PCR are also available.

Parenteral penicillin G (never give that orally in rabbits) is the treatment of choice. Other antibiotics with good activity against spirochetes include **tetracyclines and chloramphenicol.** All contact rabbits should be treated.

Perineal Impaction in Guinea Pigs

One nasty and stinky disease of guinea pigs is perineal impaction. It is **not a rectal impaction** (contrary to what you may find in some sources). Older boars develop these kinds of extra skin folds over the perineum where perineal glands are located and form kind of a sac, sometimes called pseudocloaca or perineal sac. In some cases, there can be a large accumulation of nasty stuff including **fecal matter and thick sebaceous secretions and smegma.** Boars also use a process of perineal dragging for marking using their perineal glands. There are some debates over whether this accumulation may be normal in some cases. Anyhow, when the impaction is significant, it can lead to fecal and urinary obstruction. The treatment is fairly straightforward and consists of delicately **removing the nastiness from the perineal sac.** Please, please **wear gloves**, that thing is nasty and smells like %#\$@&. You may **flush the area with diluted chlorhexidine** after. In severe cases,

guinea pigs may benefit from **antibiotic therapy and analgesics.** Husbandry may need to be corrected, especially the substrate.



Penile Diseases in Chinchillas

Chinchillas have a long penis and are very fluffy. As luck would have it, this unfortunate combination results in them being prone to accumulating hair at the base of the glans penis within the prepuce. In some cases it is relatively benign, but in some other cases, it may lead to **paraphimosis** (unable to retract penis within prepuce) and **balanoposthitis** (inflammation of the glans penis and prepuce).



It is common, so you need to inspect chinchillas' penises during physical examination. The treatment is easy but you need to be careful not to cause iatrogenic trauma in case of significant swelling. You may have to use some petroleum-based lubricant and magnification glasses to remove these fur rings safely.

Balanoposthitis may also occur for other reasons. *Pseudomonas aeruginosa* is a common bacterial culprit. Treatment is a combination of antibiotics, analgesics, and local treatment.



Preputial abscesses may also occur and will most likely require surgical lancing.



Paraphimosis is seen as a result of fur rings or balanoposthitis and may lead to anuria and self-mutilation. Treatment consists of local cleaning, using lubricant ointment, and trying to replace the penis within the pre-

puce as well as treating with antibiotics and analgesics.

Preputial Tumors in Ferrets

Preputial tumors in ferrets are very aggressive locally. They are **neoplasms of the apocrine scent glands of the prepuce. 75% of them are malignant with early local infiltration and frequent metastasis to locoregional lymph nodes and lungs.** They are almost impossible to resect if you want to get good margins. If debulked, recurrence is likely, but this can be followed up with chemotherapy and radiation therapy. Resection may involve things like penile amputation, urethrostomy, and skin flaps (needless to say that you need a good surgeon). Overall the prognosis is extremely guarded as there may already be metastasis at time of diagnosis.



The Ten Most Important Facts

1. Uterine adenocarcinoma are the most common tumors of the reproductive system in rabbits. Their prevalence is high and increases with age. Metastasis to the lungs and peritoneum are common. All rabbits should be spayed early for prevention.
2. Cystic endometrial hyperplasia is commonly diagnosed in rabbits and guinea pigs and frequency increases with age. It is typically subclinical, but can lead to vulvar discharge and large, firm uterus on palpation.
3. Ovarian cysts occur in most female guinea pigs over 2 years old. Most are serous cysts that only cause issues when too large, but a subset are follicular cysts that produce hormones and can lead to flank alopecia and other clinical signs.
4. Unspayed ferrets and ferrets with ovarian remnant syndrome may enter a state of persistent estrous (because they are induced ovulators). This can result in hyperestrogenism causing aplastic anemia and thrombocytopenia. Spaying them is the treatment of choice.
5. Mammary gland tumors are exceedingly common in rats and the majority are benign fibroadenoma. About a fourth are malignant, so histopathology should be submitted on resected tissue. As in other mammalian species, cytology is not a useful diagnostic test for these tumors. Primary prevention is by spaying early or using GnRH implants. Secondary prevention is not effective.
6. Most mammary gland tumors tend to be malignant and aggressive in rabbits and mice and half are malignant in guinea pigs and hamsters. They may also affect males in guinea pigs. A retrovirus is associated with mammary tumors in mice.
7. Rabbit syphilis is a non-zoonotic venereal spirochetosis caused by *Treponema paraluischuniculi*. Lesions are self-limiting and consist of ulcers, scabs, and inflammatory lesions of the genital areas as well as the nose and lips after auto-infection. The treatment of choice is parenteral penicillin G.
8. Older guinea pig boars have a perineal sac that can get impacted with feces, sebaceous secretions, smegma and debris. Treatment consists of cleaning and flushing.
9. Chinchillas have several penile disorders including fur rings, balanoposthitis, and paraphimosis. The penis of male chinchillas should be inspected during physical examination to screen for these fur rings.
10. Ferret preputial tumors are malignant tumors of the apocrine scent glands of the prepuce and are very aggressive with a high rate of metastasis to the locoregional lymph nodes and lungs. Surgical resection is challenging.